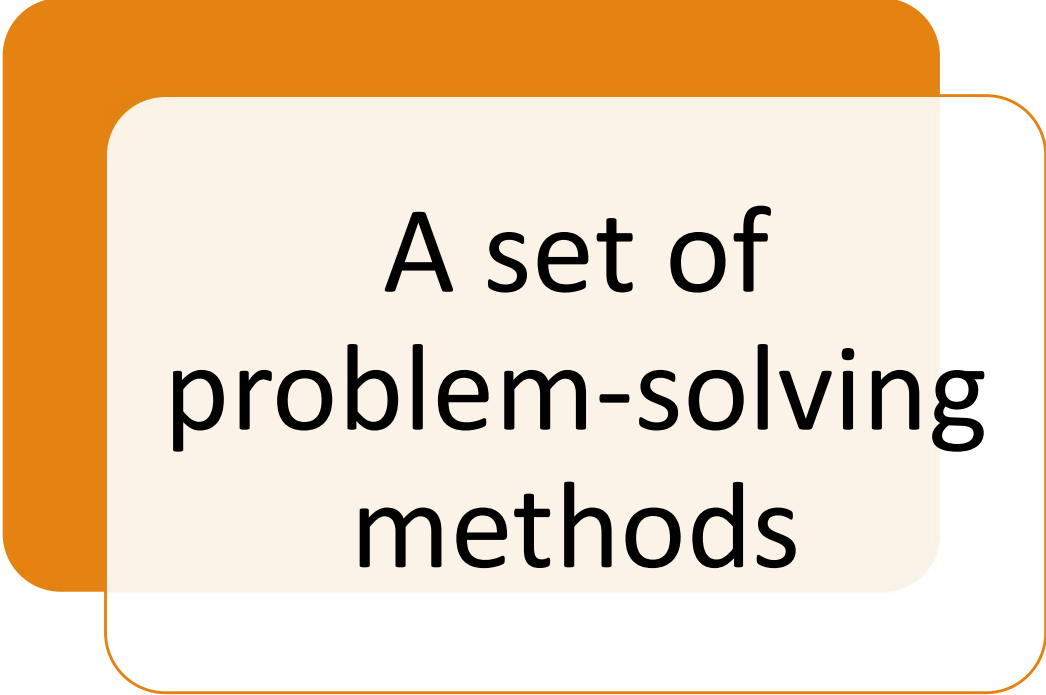


$$\begin{array}{l} \text{shoes} + \text{shoes} + \text{shoes} = 30 \\ \text{person} + \text{person} + \text{shoes} = 20 \\ \text{ice cream} + \text{ice cream} + \text{person} = 13 \\ \text{shoe} + \text{person} \times \text{ice cream} = ? \end{array}$$

Computational Thinking

A. BHULAI

What is Computational Thinking?



**A set of
problem-solving
methods**

Solving and formulating problems

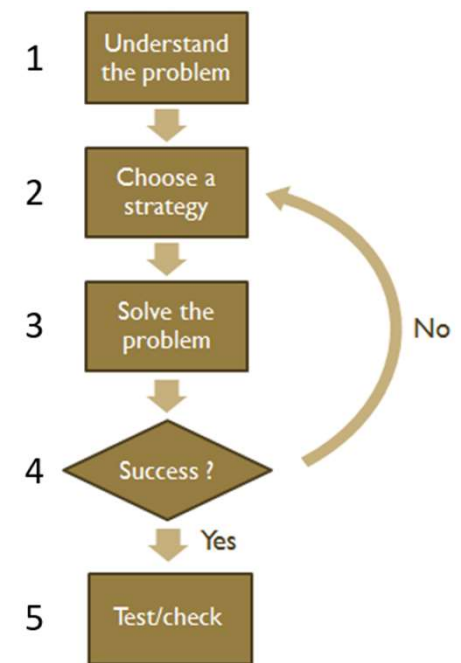
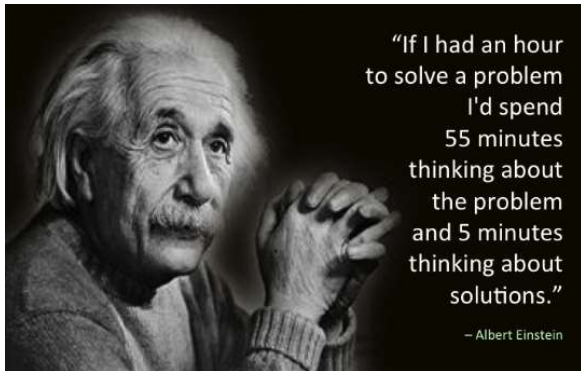
Express problems and solutions in such a way that a human and a computer can effectively execute them.



Problem formulation

Plan your steps

- Understand (e.g., decomposition)
- Possible solutions (e.g., pattern recognition)
- Develop a step-by-step solution (e.g., algorithm)
- Solve the problem (e.g., Python)



Solution strategies

- 1) The “Guess and Check” solution strategy
- 2) The “Go through all the possibilities” solution strategy
- 3) The “Divide the problem into several subproblems” solution strategy
- 4) The “Use of formulas or equations” solution strategy
- 5) The “Discover a structure or pattern” solution strategy
- 6) The “Make a model” solution strategy
- 7) The “Brute force” solution strategy
- 8) The “Divide-and-conquer” solution strategy

Procedure

- ❑ We will discuss the problem
- ❑ You work together in pairs
- ❑ 5 -10 minutes per problem
- ❑ We discuss the solution

Problem: HOMA-IR index

The HOMA-IR index is a medical indicator which is used to assess if a patient might have insulin resistance, which could lead to the development of diabetes. It can be calculated using the following formula:

$$\text{HOMA-IR} = (\text{Glucose} * \text{Insulin}) / 22.5,$$

where Glucose is the fasting glucose level in mmol/l and Insulin is the fasting insulin level in $\mu\text{IU/ml}$. The higher this value, the more resistant a person is to insulin.

Implement a Python program which takes as input the glucose level in **mg/dl** (which is the unit used in the US) and the insulin level in $\mu\text{IU/ml}$ and calculates the HOMA-IR index based on the above formula.

Hint: to convert mg/dl to mmol/l, you need to divide by 18. If you want to try out your program with realistic numbers, note that glucose levels between 70-125 mg/dl and insulin levels between 2-20 $\mu\text{IU/ml}$ are usually considered normal¹.

¹ <https://labs.selfdecode.com/blog/fasting-insulin-test/> and <https://labs.selfdecode.com/blog/fasting-blood-glucose-test/>

Solution: HOMA-IR index ^(1/2)

Solution strategy: dividing the problem into subproblems, using formulas.

STEP-BY-STEP:

- Ask user for glucose level in mg/dl.
- Ask user for insulin level in μ U/ml.
- Convert glucose level to mmol/l by dividing by 18.
- Calculate HOMA-IR using formula.
- Print HOMA-IR in message for user.

Problem:

The HOMA-IR index is a medical indicator which is used to assess if a patient might have insulin resistance, which could lead to the development of diabetes. It can be calculated using the following formula:

$$\text{HOMA-IR} = (\text{Glucose} * \text{Insulin}) / 22.5,$$

where Glucose is the fasting glucose level in mmol/l and Insulin is the fasting insulin level in μ U/ml. The higher this value, the more resistant a person is to insulin.

Task

Implement a Python program which takes as input the glucose level in **mg/dl** (which is the unit used in the US) and the insulin level in μ U/ml and calculates the HOMA-IR index based on the above formula.

Solution: HOMA-IR index – Python code (2/2)

Code

```
1 glucose_US = float(input("Please give glucose level in mg/dl."))
2 insulin = float(input("Please give insulin level in uIU/ml."))
3 glucose = glucose_US / 18
4 homa = (glucose * insulin) / 22.5
5 print("Your HOMA-IR is {fhoma:.2f}.".format(fhoma = homa))
6
```

**THANK YOU FOR YOUR ATTENTION,
YOU CAN CLAP NOW**



**IF YOU HAVE ANY QUESTIONS, PLEASE
ASK MY FRIEND GOOGLE**

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